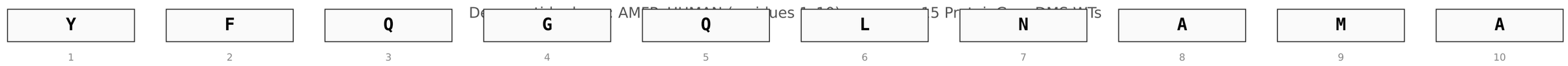
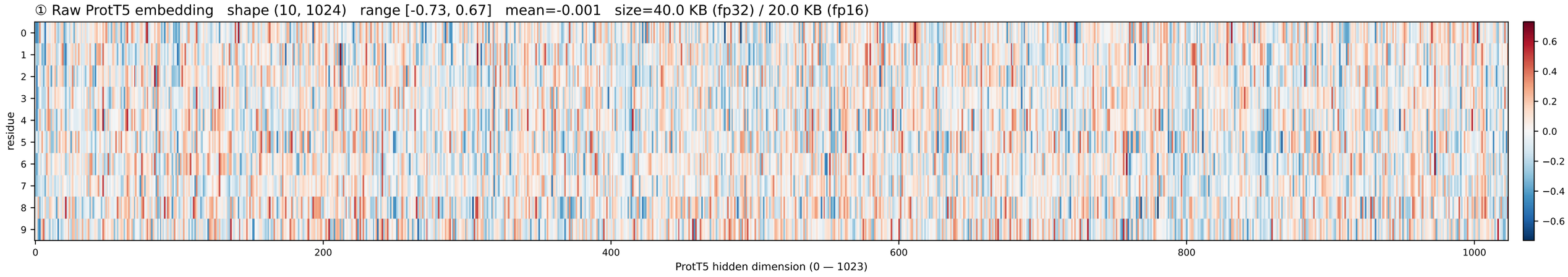
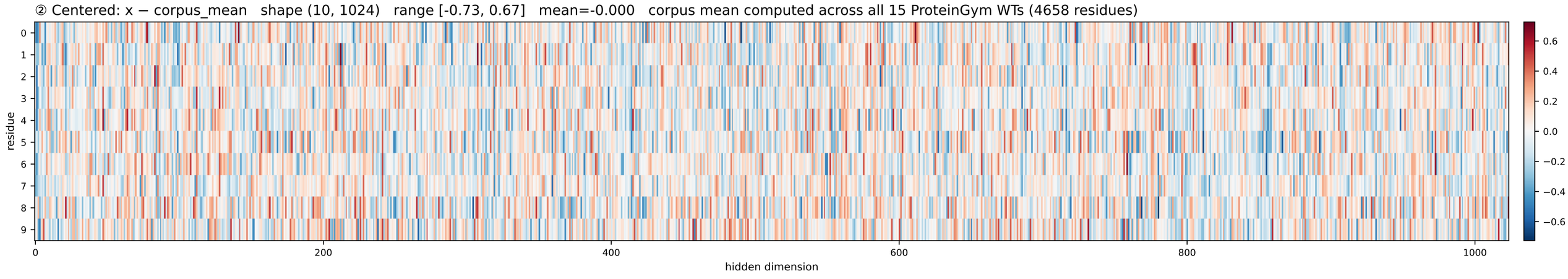
OneEmbeddingCodec — sequence → embedding → OE → DCT-K4 protein vector



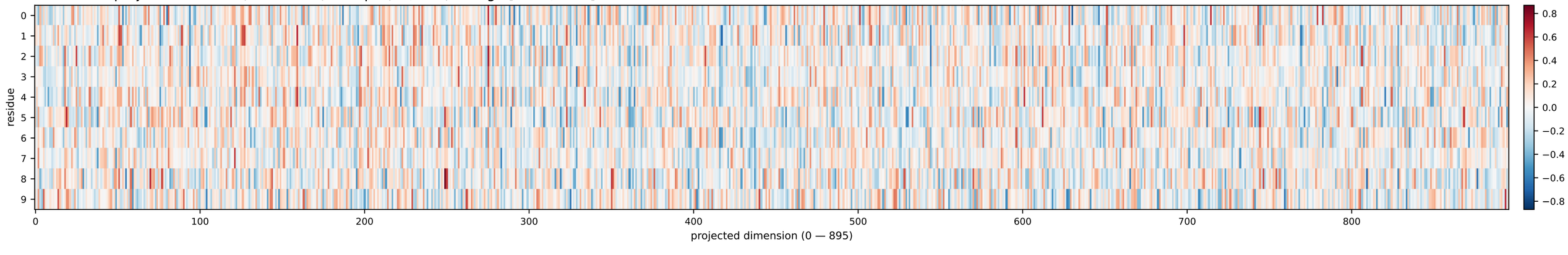
① Raw ProtT5 embedding shape (10, 1024) range [-0.73, 0.67] mean=-0.001 size=40.0 KB (fp32) / 20.0 KB (fp16)



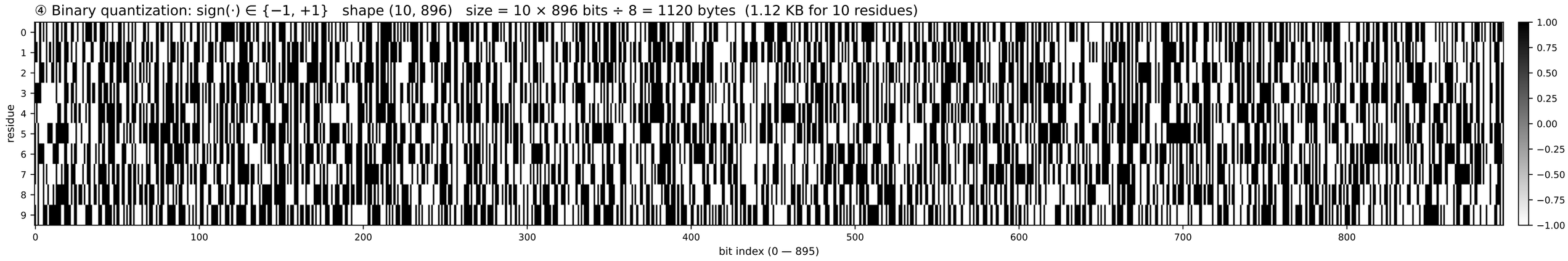
② Centered: x − corpus_mean shape (10, 1024) range [-0.73, 0.67] mean=-0.000 corpus mean computed across all 15 ProteinGym WTs (4658 residues)



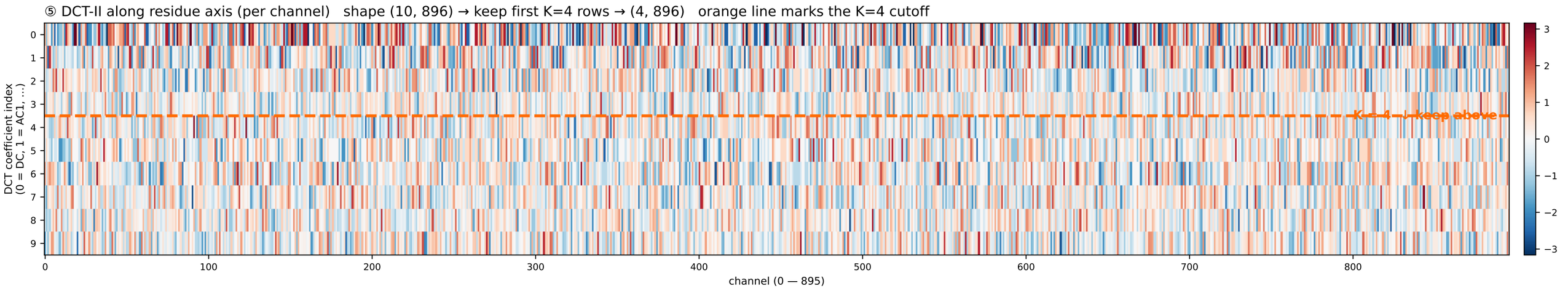
③ Random projection R ∈ ℝ^(1024×896) shape (10, 896) range [-0.75, 0.87] R is a fixed Gaussian random matrix, scale 1/√896, seed=42



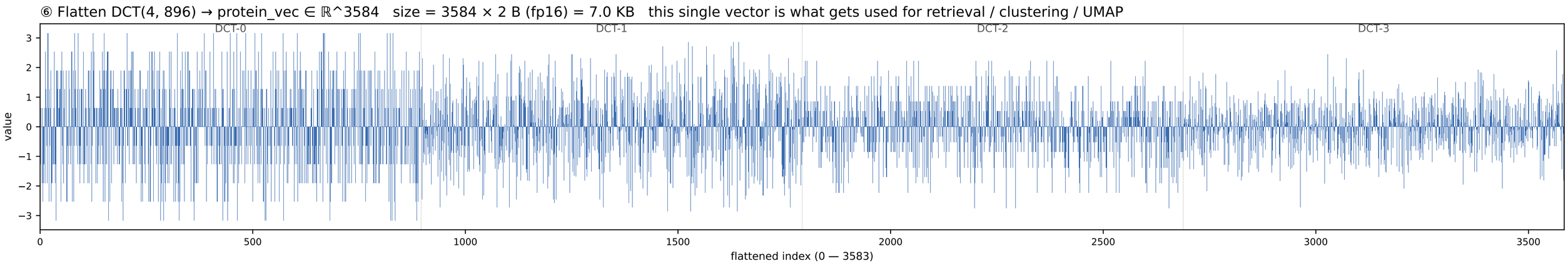
④ Binary quantization: sign(·) ∈ {−1, +1} shape (10, 896) size = 10 × 896 bits ÷ 8 = 1120 bytes (1.12 KB for 10 residues)



⑤ DCT-II along residue axis (per channel) shape (10, 896) → keep first K=4 rows → (4, 896) orange line marks the K=4 cutoff



⑥ Flatten DCT(4, 896) → protein_vec ∈ ℝ^3584 size = 3584 × 2 B (fp16) = 7.0 KB this single vector is what gets used for retrieval / clustering / UMAP




```
Per-residue payload: 10 × 896 bits = 1120 bytes (vs 40960 bytes for fp32, 20480 bytes for fp16) →  37× compression on raw fp32, 18× on fp16
Plus a single 3584-d protein vector (7.0 KB) for retrieval / clustering / UMAP — this is what appears in benchmarks like SCOPe Ret@1.
Total stored per protein:   per-residue (1120 B) + protein_vec (7168 B) = 8288 bytes for this 10-residue slice (7.2 KB at L=10; scales linearly per residue).
```